

Introduction to Reinforcement Learning

Reinforcement learning is a branch of machine learning where an agent learns by interacting with an environment. The agent takes actions and receives rewards or penalties based on those actions.

Key Concepts

1. Agent

The learner or decision maker.

2. Environment

The world the agent interacts with.

3. State

A representation of the current situation.

4. Action

A decision made by the agent.

5. Reward

Feedback received after taking an action.

Q-Learning

Q-learning is a reinforcement learning algorithm that learns the value of taking a particular action in a particular state. It uses a Q-table to estimate future rewards.

Policy Gradients

Policy gradient methods directly optimize the policy instead of estimating value functions. These methods are often used in deep reinforcement learning.

Applications

Reinforcement learning is used in:

- robotics
- game playing
- recommendation systems
- autonomous vehicles

Conclusion

Reinforcement learning allows systems to learn through experience and improve decision making over time.